AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	NYSE Data Center, NJ -- station KCDW, America/New_York
Building	OSM 472761713 NYSE Data Center
Facade gap	not applicable -- one building, no second facade
Weather record	43,594 real hourly records from KCDW
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-04-30 (chatter)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 11 of 24 hours with 1 mode change. That is 9 more than the reactive incumbent, at 0 unsafe hours against its 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 6 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	11 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	2 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
	the incumbent broke its own switch budget 1 time(s) to stay safe
	6 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dry-bulb	22.780	21.0	22.780	0.797
01	mechanical	no	dry-bulb	21.945	21.0	21.110	0.714

02	mechanical	no	dry-bulb	21.360	21.0	19.440	0.870
03	mechanical	yes	switch budget	20.830	21.0	18.330	0.817
04	mechanical	yes	switch budget	19.725	21.0	17.780	0.757
05	mechanical	yes	switch budget	18.610	21.0	17.220	0.668
06	mechanical	yes	switch budget	18.055	21.0	16.670	0.729
07	mechanical	yes	switch budget	18.890	21.0	16.670	1.217
08	mechanical	yes	switch budget	20.000	21.0	17.220	1.496
09	mechanical	no	dry-bulb	21.115	21.0	17.780	1.956
10	mechanical	no	dry-bulb	21.670	21.0	18.890	1.791
11	mechanical	no	dry-bulb	21.390	21.0	18.890	1.479
12	mechanical	no	dry-bulb	21.115	21.0	18.890	1.246
13	FREE	yes	-	19.725	21.0	16.110	1.051
14	FREE	yes	-	20.550	21.0	18.330	1.115
15	FREE	yes	-	19.445	21.0	17.220	0.981
16	FREE	yes	-	18.610	21.0	19.440	0.904
17	FREE	yes	-	19.165	21.0	18.890	1.119
18	FREE	yes	-	17.220	21.0	17.220	1.012
19	FREE	yes	-	16.665	21.0	14.440	1.039
20	FREE	yes	-	15.305	21.0	12.220	1.202
21	FREE	yes	-	14.165	21.0	11.110	1.353
22	FREE	yes	-	11.940	21.0	9.440	1.143
23	FREE	yes	-	10.555	21.0	8.890	0.935

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 22.780 C against a 21.0 C limit, so it fails by 1.780 C. It would take a limit 1.780 C higher, or a bound 1.780 C tighter, to change this hour.

01:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.945 C against a 21.0 C limit, so it fails by 0.945 C. It would take a limit 0.945 C higher, or a bound 0.945 C tighter, to change this hour.

02:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.360 C against a 21.0 C limit, so it fails by 0.360 C. It would take a limit 0.360 C higher, or a bound 0.360 C tighter, to change this hour.

03:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.830 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.725 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.610 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes

per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.055 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.890 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.000 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.115 C against a 21.0 C limit, so it fails by 0.115 C. It would take a limit 0.115 C higher, or a bound 0.115 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.670 C against a 21.0 C limit, so it fails by 0.670 C. It would take a limit 0.670 C higher, or a bound 0.670 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.390 C against a 21.0 C limit, so it fails by 0.390 C. It would take a limit 0.390 C higher, or a bound 0.390 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.115 C against a 21.0 C limit, so it fails by 0.115 C. It would take a limit 0.115 C higher, or a bound 0.115 C tighter, to change this hour.

13:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.725 C, which is 1.275 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.051 C, and the margin is measured, not chosen: 1.051 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

14:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.550 C, which is 0.450 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.115 C, and the margin is measured, not chosen: 1.115 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

15:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.445 C, which is 1.555 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.981 C, and the margin is measured, not chosen: 0.981 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.610 C, which is 2.390 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.904 C, and the margin is measured, not chosen: 0.904 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.165 C, which is 1.835 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.118 C, and the margin is measured, not chosen: 1.118 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.220 C, which is 3.780 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.012 C, and the margin is measured, not chosen: 1.012 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.665 C, which is 4.335 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.039 C, and the margin is measured, not chosen: 1.039 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.305 C, which is 5.695 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.202 C, and the margin is measured, not chosen: 1.202 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.165 C, which is 6.835 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.353 C, and the margin is measured, not chosen: 1.353 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.940 C, which is 9.060 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.142 C, and the margin is measured, not chosen: 1.142 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.555 C, which is 10.445 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.935 C, and the margin is measured, not chosen: 0.935 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

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